

- π^* is the AIXI policy.
- We want to see if training a student on short horizon trajectories generated by π^* will lead to a student policy that generalizes to longer horizons.

Simplest case:

- Sample independent $a_{1:t}^1, \dots, a_{1:t}^N \sim \pi^*$ for some small t .
- $w(\pi_i | a_{1:t}^1, \dots, a_{1:t}^N) \propto w(\pi_i) \pi_i(a_{1:t}^1) \cdots \pi_i(a_{1:t}^N)$
- $\zeta_N = \sum_i w(\pi_i | a_{1:t}^1, \dots, a_{1:t}^N) \pi_i$ is the student policy.
- As $N \rightarrow \infty$, $\zeta_N \rightarrow \zeta$.
- Can we show convergence in TV of ζ to π^* as $t \rightarrow \infty$?
- Can we then show convergence of ζ to optimality in terms of the value function $V^\zeta(\cdot | h_{<t})$?

Convergence rate:

- In general, convergence of ζ to π^* can be very slow. We can have a large mass of hypotheses that are consistent with π^* only up to time t , and then sharply deviate.
- Deviating from π^* at time t incurs a complexity cost of roughly $K(t)$, which goes to infinity as $t \rightarrow \infty$. So policies that make a sharp deviation at time t will have a prior weight of at most $2^{-K(t)}$.
- More generally, the behavior of stopping complexity would be relevant.
- Under the simplicity prior, can we show some convergence rate of ζ to π^* ?

Considering sample size N :

- Let's consider the rate of convergence w.r.t. N .
- Suppose t is sampled from some distribution, say the exponential distribution.
- How fast does ζ converge to π^* as $N \rightarrow \infty$?

Unrealizability:

- How do the above questions change if π^* is not in the hypothesis class? e.g. AIXI is not computable.
- Although π^* is not computable, $\pi_{1:t}^*$ is computable. How hard is it to describe $\pi_{1:t}^*$ without a halting oracle? i.e. What is $K(\pi_{1:t}^*)$ as a function of t ?

Reinforcement learning:

- Above, we considered supervised learning with AIXI as the teacher, because it's simpler to think about than RL. The only difference with RL is that RL doesn't dictate how to tie-break actions.
- How do the above questions change if we use RL instead of supervised learning?
- e.g. In the simplest case, we can have an environment μ , and we define $w_t(\pi_i) = w(\pi_i) \mathbb{I}[\pi_i \in \arg \max_{\pi} V_{\mu_{1:t}}^\pi]$ with finite horizon t .